PCB Design Report

Adafruit Circuit Playground Express (Round 1, Problem 2)

Submitted By-

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EDA Tool: **EasyEDA**

Contest: iTest Bangladesh PCB Design Contest 2026, Round One

Date: 06/10/2026

1. Introduction

This report describes the design of a compact interactive embedded controller inspired by the Adafruit Circuit Playground Express. The board combines an ATSAMD21G18 microcontroller, a USB interface, LiPo battery input, ten NeoPixel LEDs, motion, sound, light and temperature sensors, a speaker, IR link, buttons and capacitive touch pads.

2. Component List (BOM)

Table 1 lists all components with designator, value, package and source. Parts from the EasyEDA library are marked "Library"; parts built from datasheet dimensions are marked "Custom".

Designator	Component	Value	Package	Source
U$5	ATSAMD21G18MU MCU	-	QFN-48	[Library/Custom]
U4	GD25Q16C SPI flash	2MB	[SOIC-8/WSON-8]	[Library]
IC4	LIS3DH accelerometer	-	LGA-16	[Library/Custom]
IC2	PAM8301 audio amplifier	-	SOT23-6	[Library]
U1	VSOP38338 IR receiver	-	[package]	Custom
U3	AP2112K-3.3 LDO	3.3V	SOT23-5	[Library]
Q1	MMBT2222 NPN transistor	-	SOT23	[Library]
D1, D2	MBR120 Schottky diode	-	[SOD-123]	[Library]
U2	ALS-PT19-315C light sensor	-	[package]	Custom
IC1	SPH0645LM4H-B I2S microphone	-	[package]	Custom
NeoPixel x10	WS2812B	-	5050	Custom/Library
THM1	NTC thermistor	10k 1%	[0603]	[Library]
SP1	Surface-mount speaker	7.5mm / 8R	[package]	Custom
D3, D5	PIN photodiode, IR LED	-	[package]	[Library]
F1	Resettable fuse	6V, 500mA	[1206]	[Library]
CN1, BATT	Micro-B USB, JST-PH 2-pin	-	[package]	[Library]
RESET, SW1, SW2	KMR2, EVQQ tactile switches	-	[package]	Custom
SWITCH.1, .2	EG1390 slide switch	-	[package]	Custom
L, PWR	Red and green LEDs	-	[0603]	[Library]
R1-R10, C1-C10	Resistors, capacitors	per schematic	[0603/0805]	[Library]

Table 1: Bill of materials

3. Schematic

The schematic is divided into five sections: Power Supply, ATSAMD21 Microcontroller, NeoPixels and Sensors, Speaker and IR, and Capacitive Touch pads. Net labels are used for all signals so that connections stay clear and readable.

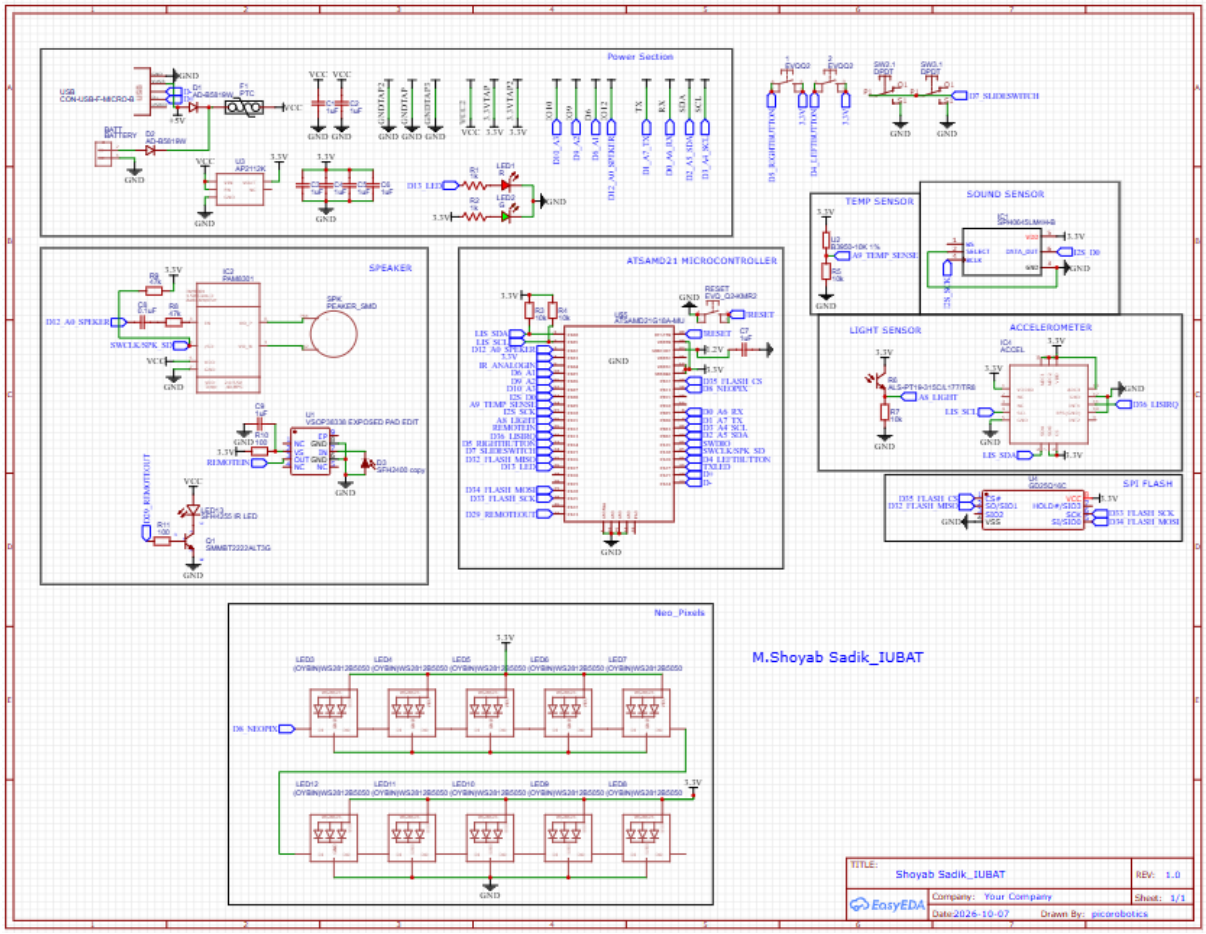


Fig1: Schematic of the PCB

4. PCB Design Decisions

The board is [Width: 80mm & Height: 39.169mm] with [2] layers. The ten NeoPixels are placed in a ring so the chain follows DO to DI. The USB and battery connectors sit at the board edge. Decoupling capacitors are placed next to the ATSAMD21 supply pins, and the 1.2 V core capacitor C10 sits next to VDDCORE. The sensors are placed away from the LEDs, speaker and power section.

Requirement	How it was met
Decoupling close to supply pins	[distance of C10 and other caps from U$5 pins]
USB/programming near edge	[CN1, BATT, F1, D1, D2 position]
Sensors away from noisy power	[IC4, THM1, U2, IC1 distance from U3, IC2, SP1]
Short, organised I2C	[LIS_SDA and LIS_SCL length to IC4]
Accessible connectors and headers	[CN1, BATT, TP pads on edge]
Clear silkscreen	[designators, connector labels, pad names]

Table 3: Requirements checklist

5. Design Rules

Signal traces are 0.25 mm. Clearance is 0.3 mm and vias are 0.35 mm. A GND copper pour is used on both layers with stitching vias. The capacitive touch pads are exposed copper with a solder mask opening.

6. PCB Layout Views

The figure2 show the 2D top layer. Figures 3 and 4 show the PCB Design view and 3D view respectively.

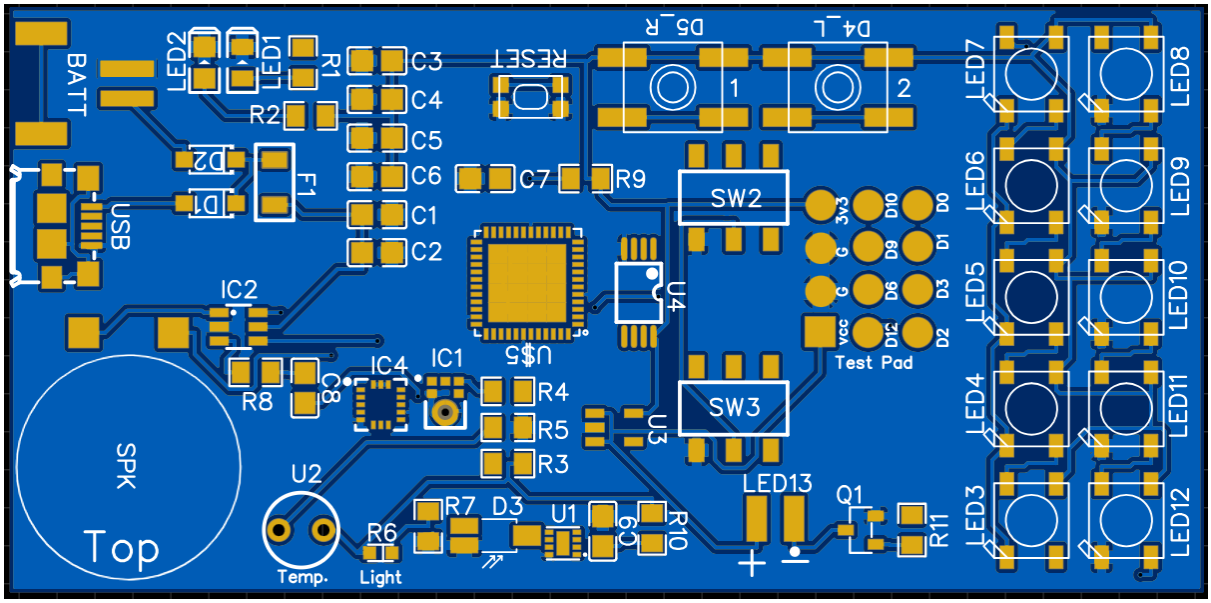


Fig2: Top View

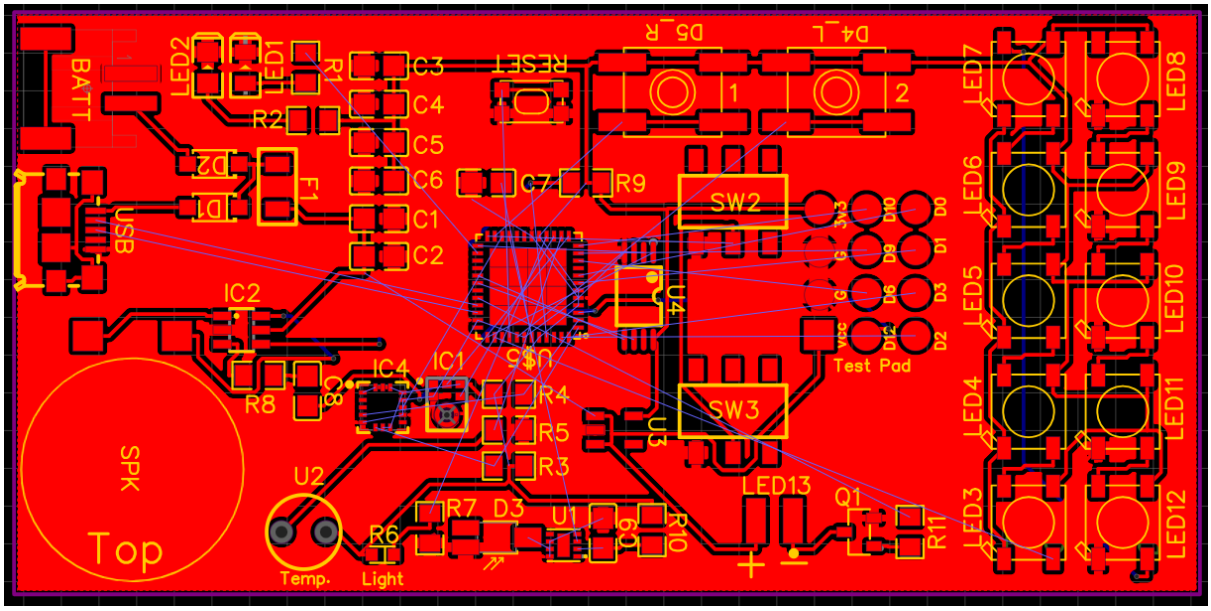


Fig3: PCB Design View

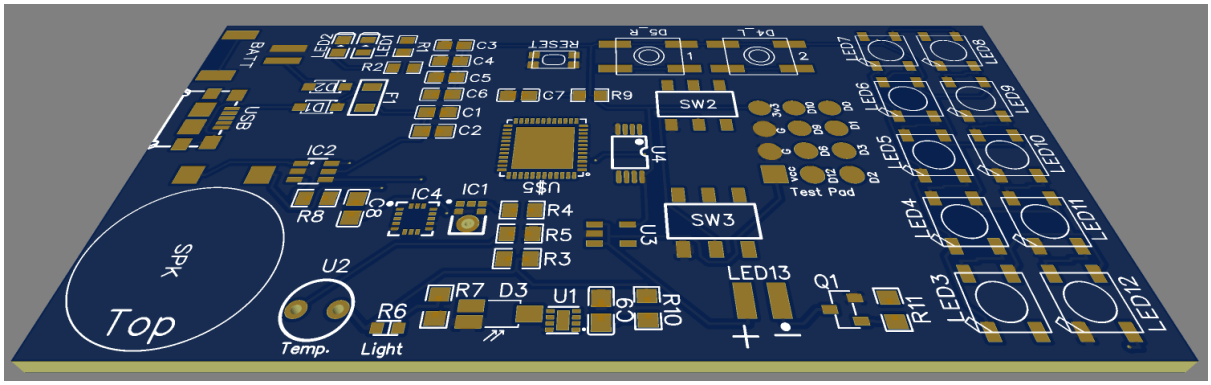


Fig4: 3D View of the PCB

7. Conclusion

The design meets the stated requirements: correct footprints, close decoupling, USB near the edge, short I2C routing, accessible connectors, clear silkscreen and a clean DRC.

8. Files Submitted

- Shoyab_Sadik_IUBAT schematic file
- Shoyab_Sadik_IUBAT board file
- Shoyab_Sadik_IUBAT pdf (all layers, one per page)
- Shoyab_Sadik_IUBAT report (this document)